# Introduction to Autonomous Vehicles

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1

## Outline Introduction to Autonomous Vehicles

- Perception
  - Machine Learning Workflow
  - Sensor and Camera Calibration
  - Neuronal Networks
  - Image Classification
  - 2D Object detection
- Sensor Fusion
  - Sensor Fusion and Perception
  - Lidar Sensor
  - 3D Object Detection
  - Kalman Filters and Extended Kalman filters
  - Multi Object Tracking
- Localization
  - Markov Localization
  - Scan Matching Algoritms 2D and 3D
- **Decision Making and Planning**
  - **Behavioral Planning**
  - Trajectory Generation
  - Motion Planning
- Control
  - Proportional-Integral-Derivative (PID) Control
  - Model Predictive Control (MPC)
- Human Factors



2

# Decision Making and Planning

Allow self-driving vehicles to operate safely, efficiently, and predictably in a wide range of traffic scenarios.

Based on the sensors information the vehicle must make a series of decisions and plans to navigate through traffic and reach its destination



# Decision Making and Planning

Effective decision making and planning involve adapting to changing circumstances in real-time considering:

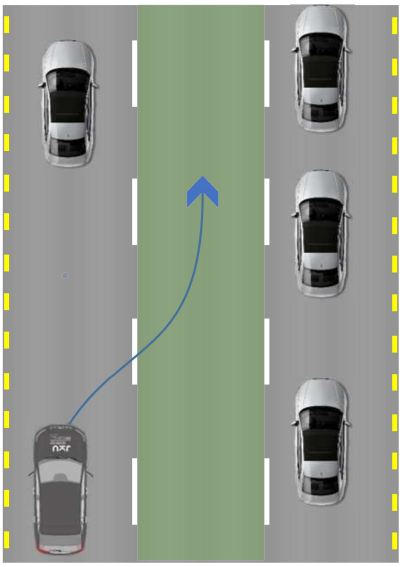
- vehicle's speed and position
- behavior of other vehicles on the road
- road and weather conditions
- passenger preferences

# Behavior Planning

**Introduction:**
A behavior planning system plans the set of high-level driving actions, or maneuvers to safely achieve driving under various situations.

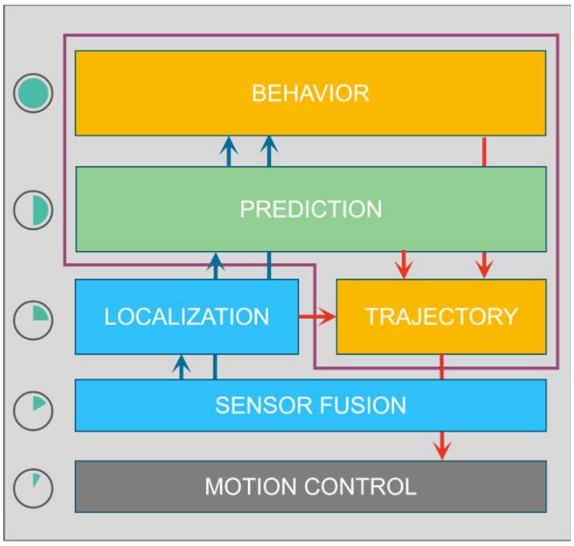
**Behavior planner considers:**
- Rules of the road
- Objects around the ego vehicle
- Sometimes the route to the destination

Planned behavior must be safe and as efficient as possible.





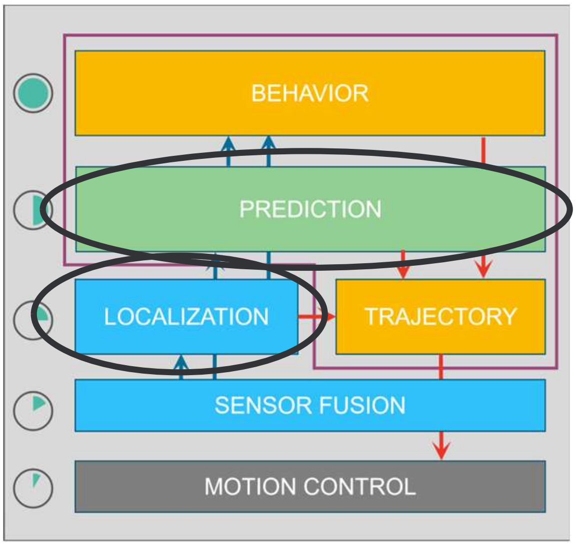
# Path Behavior Planning



## Path Behavior Planning

**Input of Behavior Planner:**
- Road map
- Ego vehicle information (route to the destination, localization, speed, etc.)

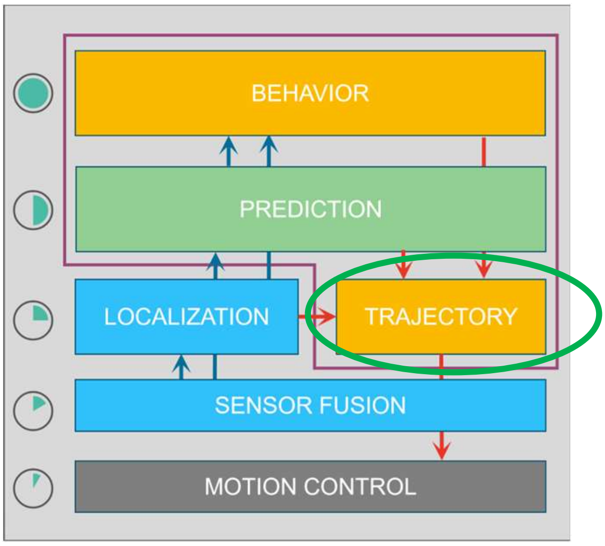
- Perception Information:
  - ➢ Static objects
    e.g. road signs; traffic lights
  - ➢ Dynamic objects
    e.g. prediction of position and future movement; estimated time to collision






## Path Behavior Planning

**Output of Behavior Planner:**
- Driving maneuvers to be executed



# Behavior Planning

## Example driving maneuvers

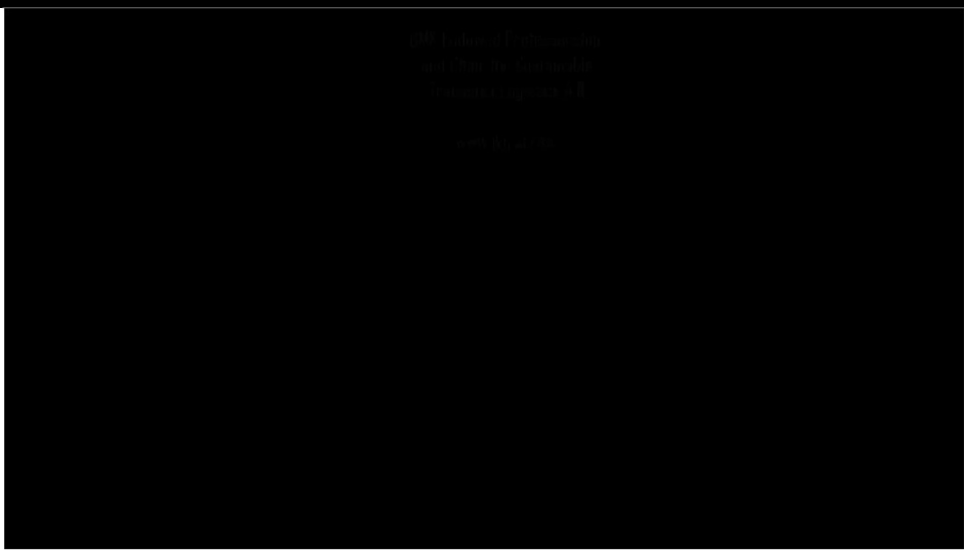
- **Speed keeping:** maintain current speed
- **Lane keeping:** maintains its position within a lane
- **Lane changing:** moves from one lane to another when necessary
- **Merging:** merge into another lane of traffic
- **Overtaking:** safely overtake slower-moving vehicles on the road
- **Avoidance:** avoid obstacles, pedestrians, and other vehicles on the road.
- **Speed adjustment:** the car can adjust its speed to match the speed limit or traffic conditions
- **Leader following:** match the speed of the leading vehicle and maintain a safe distance
- **Stopping:** decelerate to stop; emergency stopping
- **...**





# Behavior Planning

Speed keeping
Lane keeping
Lane changing



"Interaction of Autonomous and Manually-Controlled Vehicles: Implementation of a Road User Communication Service", Smirnov, N., Tschernuth, S., Morales-Alvarez W., and Olaverri-Monreal, C.(2022) IEEE Intelligent Vehicle Symposium 2022, Aachen, Germany.

## Behavior Planning

Avoidance
Speed adjustment
Stopping



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11

## Behavior Planning

Merging


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12

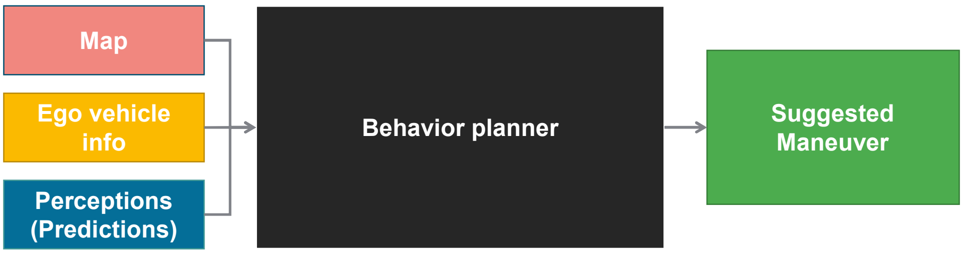
## Behavior Planning

Overtaking

Olaverri-Monreal, C., Gomes, P., Fernandes, R., Vieira, F., Ferreira, M. (2010) "The See-Through System: A VANET-Enabled Assistant for Overtaking Maneuvers" Proceedings 2010 IEEE Intelligent Vehicles Symposium IV. San Diego, CA, USA. pp. 123–128.



## Behavior Planning




**RESPONSBILITIES:**
Suggest "states" (maneuvers) which are:
- Feasible
- Safe
- Legal
- Efficient

Execution Details

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**Control System**